Forecasting Atmospheric CO_2 with SARIMA and Explaining the Drivers with SHAP

A data science whitepaper on seasonal time-series modelling, uncertainty quantification and model interpretability

Task 4 - Mini Project: Time-Series Forecasting Model & Advanced Model Interpretability Dashboard

1. Executive Summary

This paper fits a seasonal ARIMA model to 526 monthly observations of atmospheric CO_2 measured at Mauna Loa (1958-03-01 to 2001-12-01), tests the series for stationarity, removes trend and seasonality through differencing, and produces forecasts with 95% confidence intervals. The selected specification, **SARIMA(0,1,1)x(1,1,1,12)**, was chosen by an AIC search over 36 candidate models. On a 24-month hold-out the model reached an RMSE of **0.431 ppm** (MAE 0.357, MAPE 0.096%) for the multi-step forecast, and its 95% intervals covered **100%** of the realised values. A second, machine-learning model built on lagged seasonal differences was explained with SHAP to quantify exactly how much each input contributes to a forecast. The most influential feature was **sdiff_lag1** (49.0% of total attribution), followed by sdiff_roll6 and sdiff_lag12.

2. Introduction and Objectives

Accurate forecasts are only useful when decision makers can trust them. Classical statistical models such as SARIMA give calibrated uncertainty and interpretable coefficients, whereas flexible machine-learning models often reach good accuracy but behave as black boxes. This project combines both views. The objectives are: (i) load a historical sequential climate dataset; (ii) diagnose non-stationarity and apply a seasonal differencing pipeline; (iii) train a SARIMA forecaster that predicts future values with confidence intervals; (iv) wrap a model in an interpretability layer (SHAP) that measures exact feature-importance weights; and (v) report the findings in this whitepaper and a one-page dashboard.

3. Data

The data are the Keeling/Scripps Mauna Loa weekly atmospheric CO_2 record distributed with the *statsmodels* library. The weekly series contains 59 missing values; it was aggregated to monthly means and remaining gaps were filled by time-based linear interpolation. Values range from 313.40 to 373.80 ppm (mean 339.62 ppm). Figure 1 shows the long-run upward trend together with a regular within-year cycle caused by seasonal vegetation uptake in the northern hemisphere.

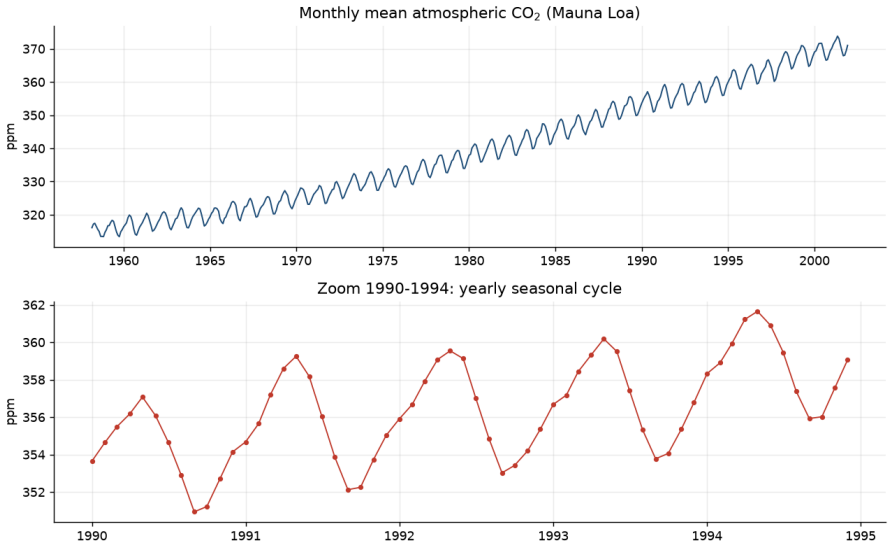


Figure 1 - Monthly CO_2 series (top) and a five-year zoom showing the annual cycle (bottom).

An STL decomposition (Figure 2) separates the series into trend, seasonal and remainder components. The trend rises by about 56.23 ppm over the sample, the seasonal cycle has a peak-to-trough amplitude of about 6.71 ppm, and the strength of trend and seasonality are 1.000 and 0.984 respectively (1 = perfectly deterministic). Both

structures must be handled by the model.

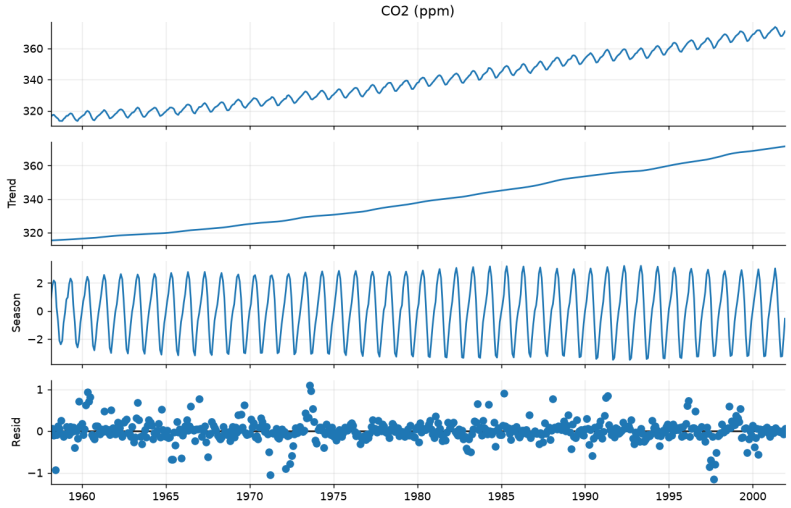


Figure 2 - STL decomposition (period = 12) into trend, seasonal and remainder.

4. Stationarity Analysis and Seasonal Differencing

SARIMA assumes that, after differencing, the series has a constant mean and variance structure. Two complementary tests were used: the Augmented Dickey-Fuller (ADF) test, whose null hypothesis is a unit root (non-stationarity), and the KPSS test, whose null hypothesis is stationarity. A series is declared stationary only when ADF rejects ($p < 0.05$) *and* KPSS does not reject ($p > 0.05$). Using both tests protects against the low power of either test alone.

Series	ADF stat	ADF p	KPSS stat	KPSS p	Verdict
Raw series	2.234	0.999	3.369	0.010	Non-stationary
First difference (d=1)	-4.753	0.000	0.300	0.100	Stationary
Seasonal difference (D=1, s=12)	-4.291	0.001	1.183	0.010	Non-stationary
d=1 and D=1 (s=12)	-8.401	0.000	0.020	0.100	Stationary

Table 1 - Stationarity tests after each transformation. The raw series is non-stationary (ADF p = 0.999, KPSS p = 0.010); after one regular and one seasonal (lag-12) difference the verdict is **stationary** (ADF p = 0.000, KPSS p = 0.100). KPSS p-values are truncated by the test to the interval [0.01, 0.10].

The first difference removes the stochastic trend, and the seasonal difference $y_t - y_{t-12}$ removes the annual cycle (Figure 3). Their combination motivates d = 1 and D = 1 with seasonal period s = 12 in the SARIMA specification. The ACF/PACF of the doubly differenced series (saved as fig4_acf_pacf.png) show short-lag and seasonal-lag structure that guides the candidate range for the autoregressive (p, P) and moving-average (q, Q) orders.

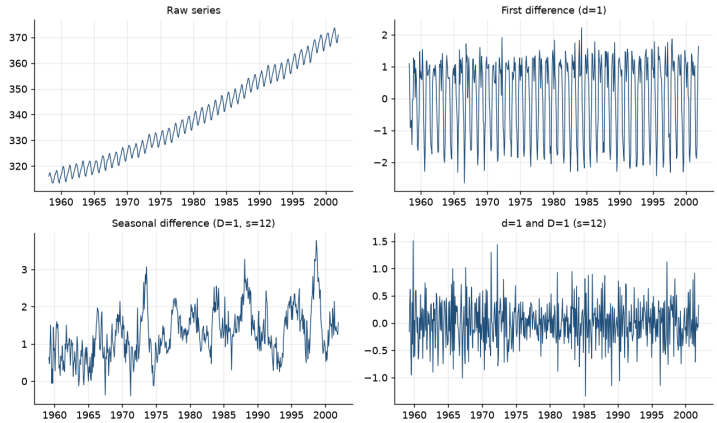


Figure 3 - Effect of regular and seasonal differencing.

5. SARIMA Model and Forecast

A SARIMA(p,d,q)x(P,D,Q)$_s$ model applies the differencing above and models the result with autoregressive and moving-average terms at both the regular and seasonal lags. Parameters are estimated by maximum likelihood on a Kalman-filter state-space form. Holding out the final 24 months, a grid search over p, q in {0,1,2} and P, Q in {0,1} (d = D = 1, s = 12) was run on the training window and ranked by the Akaike information criterion (Table 2).

Rank	Candidate model	AIC	BIC
1	SARIMA(0, 1, 1)x(1, 1, 1, 12)	207.66	224.31
2	SARIMA(1, 1, 1)x(1, 1, 1, 12)	208.16	228.97
3	SARIMA(0, 1, 2)x(1, 1, 1, 12)	209.11	229.91
4	SARIMA(2, 1, 1)x(1, 1, 1, 12)	211.56	236.54
5	SARIMA(1, 1, 2)x(1, 1, 1, 12)	211.77	236.73

Table 2 - Five best candidates by AIC on the training window.

The winner, **SARIMA(0,1,1)x(1,1,1,12)**, was used to forecast the 24-month hold-out (Figure 4). Prediction intervals come from the model's forecast error variance, which grows with the horizon. The 95% interval was on average 2.77 ppm wide and contained 100% of the actual values.

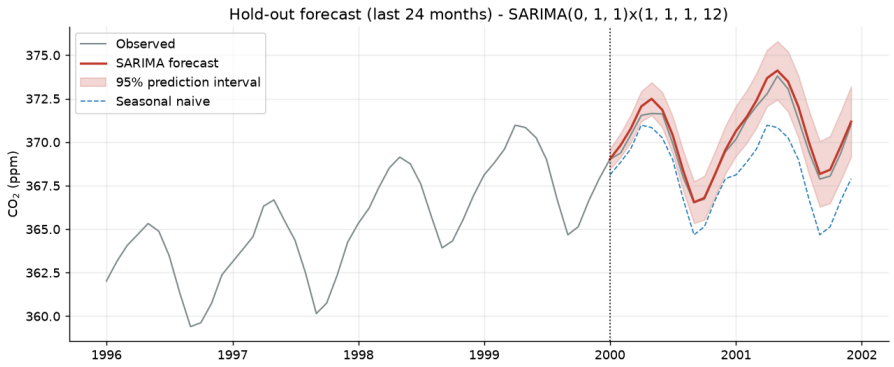


Figure 4 - Hold-out forecast with 95% prediction interval against a seasonal-naive benchmark.

5.1 Accuracy and benchmarks

Model	Horizon	MAE	RMSE	MAPE %
SARIMA	24-step	0.357	0.431	0.096
Seasonal naive	24-step	1.882	2.066	0.508
SARIMA (params frozen)	1-step	0.210	0.272	0.057
Seasonal naive	1-step	1.314	1.374	0.355
Gradient boosting + lags	1-step	0.297	0.394	0.080

Table 3 - Hold-out accuracy in ppm. Multi-step rows forecast all months from the end of training; one-step rows predict each month given all earlier observations.

For the 24-step forecast SARIMA improved on the seasonal-naive benchmark (RMSE 0.431 versus 2.066 ppm). Among one-step-ahead predictors the lowest RMSE (0.272 ppm) was achieved by SARIMA(0,1,1)x(1,1,1,12). Because the benchmark simply repeats last year's value, it cannot capture the persistent upward trend, which is why differencing-based models are the natural reference for this series.

5.2 Residual diagnostics

A well-specified model leaves residuals that resemble white noise. The Ljung-Box test at 24 lags gave p = 0.632, so there is no significant remaining autocorrelation. The Jarque-Bera normality test gave p = 0.003 and the residual standard deviation is 0.313 ppm (Figure 5).

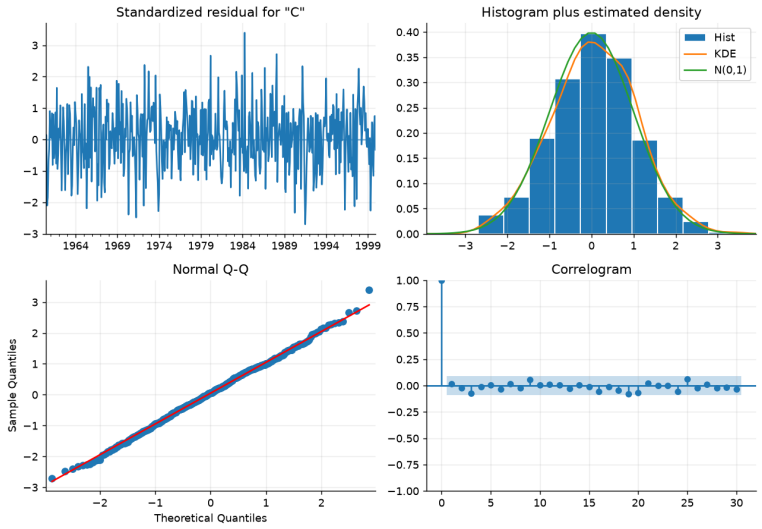


Figure 5 - Standardised residuals, histogram/density, Q-Q plot and correlogram.

5.3 Future forecast

After selecting the specification, the model was refit on the complete history and projected 36 months ahead (Figure 6). The point forecast moves from 371.97 ppm in 2002-01 to 375.70 ppm in 2004-12, where the 95% confidence interval spans 373.13 to 378.28 ppm. The widening band reflects the accumulation of uncertainty from the integrated (differenced) components.

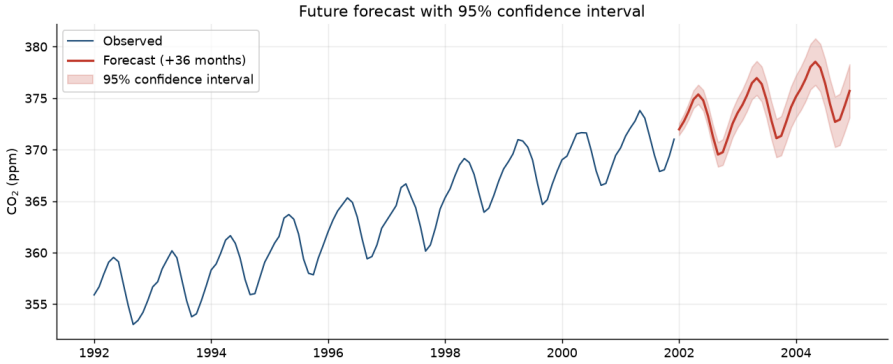


Figure 6 - Future forecast with 95% confidence interval.

6. Model Interpretability

Interpretability is addressed at two levels. SARIMA is *intrinsically* interpretable: each fitted coefficient has a direct meaning (Table 4). Because SARIMA is linear, however, SHAP and LIME - which target flexible models - are applied to a complementary tree-ensemble forecaster whose predictions can be decomposed exactly.

Term	Coefficient	Std. error	p-value
ma.L1	-0.354	0.042	0.000
ar.S.L12	-0.001	0.001	0.068
ma.S.L12	-0.852	0.029	0.000
sigma2	0.087	0.006	0.000

Table 4 - Fitted SARIMA coefficients with standard errors. Moving-average terms close to -1 indicate that shocks are largely absorbed by the differencing, a typical pattern for series that were possibly over-differenced or strongly seasonal.

6.1 SHAP on a lag-feature model

A gradient-boosting regressor (400 depth-3 trees, learning rate 0.03) predicts the seasonal difference $y_t - y_{t-12}$ from: three short lags of that difference, its 12-month lag, a 6-month rolling mean, last month's month-on-month change, and sine/cosine encodings of the calendar month. Predicting the difference rather than the level avoids the well-known inability of trees to extrapolate a trend. The forecast of the level is recovered by adding back y_{t-12}. SHAP

values (Shapley values from cooperative game theory, computed exactly for trees by TreeSHAP) distribute each prediction among the features so that the contributions plus the base value sum to the model output.

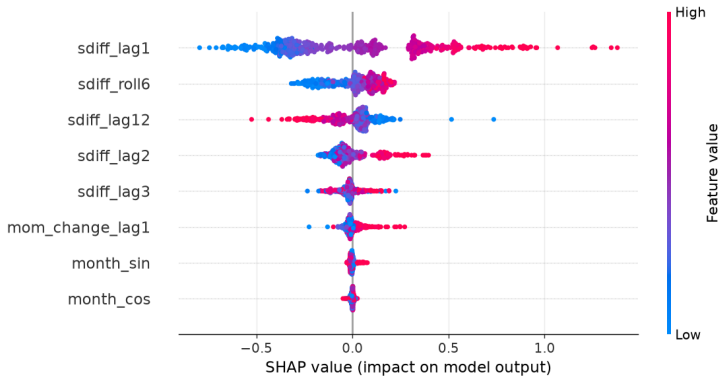


Figure 7 - SHAP beeswarm: each dot is one month; colour shows the feature value and horizontal position its effect on the prediction (ppm).

Table 5 gives the exact global importance weights. The top three features - sdiff_lag1, sdiff_roll6 and sdiff_lag12 - jointly account for 77.4% of total attribution. The ranking is consistent with the tree model's own impurity-based importances, whose top entries were sdiff_lag1, sdiff_lag2, sdiff_lag12; SHAP is preferred because it is additive, locally exact and does not favour high-cardinality features.

Rank	Feature	Mean |SHAP| (ppm)	Share of total %
1	sdiff_lag1	0.3497	49.0
2	sdiff_roll6	0.1090	15.3
3	sdiff_lag12	0.0938	13.1
4	sdiff_lag2	0.0787	11.0
5	sdiff_lag3	0.0404	5.7
6	mom_change_lag1	0.0286	4.0
7	month_sin	0.0087	1.2
8	month_cos	0.0051	0.7

Table 5 - Global feature-importance weights from SHAP (mean absolute value over all modelled months).

6.2 Local explanation

Figure 8 explains a single forecast, the month 2001-12. Starting from the average model output of 1.306 ppm, each feature pushes the predicted year-on-year change up or down until it reaches the final prediction of 1.421 ppm. Such a breakdown lets an analyst audit any individual forecast.

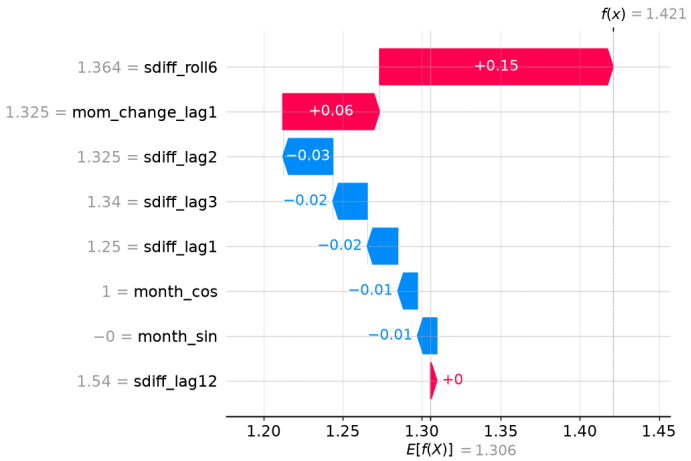


Figure 8 - SHAP waterfall plot for the most recent month.

6.3 Dashboard

Figure 9 consolidates the forecast, model comparison, SHAP ranking, residual ACF and seasonal component into one view for stakeholders.

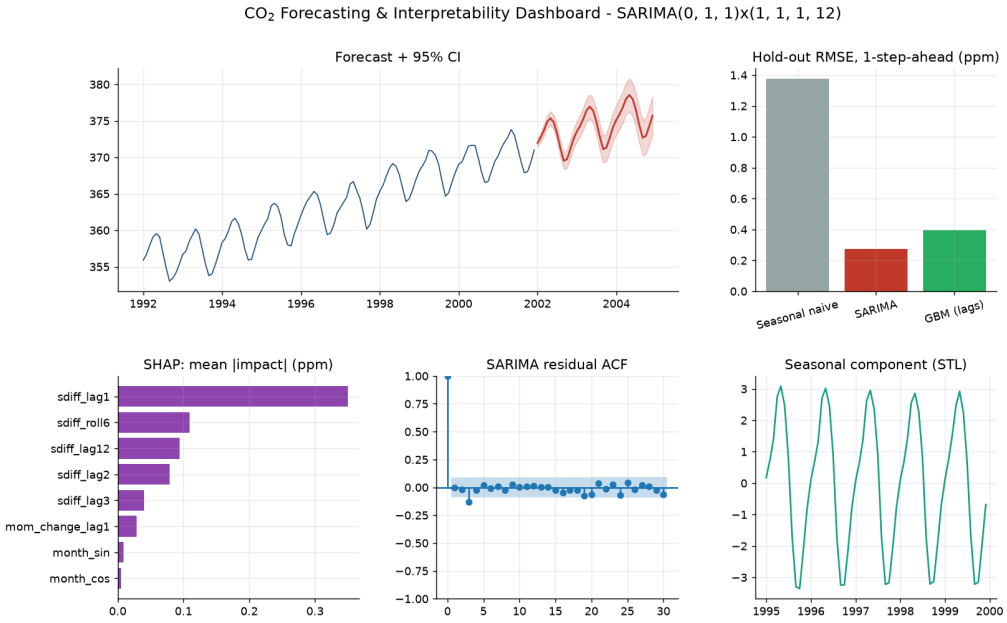


Figure 9 - Forecasting and interpretability dashboard.

7. Discussion and Limitations

Uncertainty. SARIMA intervals assume Gaussian, homoscedastic errors; the residual tests above show how well this holds. Intervals widen with horizon, so long-range forecasts should be read as scenarios, not guarantees. **Structural change.** The fitted model extrapolates the historical growth pattern and cannot anticipate policy-driven or physical regime changes. **Interpretation scope.** SHAP explains the *model*, not the physical world: a large attribution to a lag means the model relies on persistence, not that the lag causes CO_2 changes. **Data.** Series ends in 2001; applying the pipeline to newer observations is straightforward since the code is data-source agnostic once a monthly Series is provided. **Future work** includes exogenous regressors (e.g. ENSO index), Prophet as a comparator, rolling-origin cross-validation and conformal prediction intervals.

8. Conclusion

A seasonal ARIMA model selected by information criteria, preceded by a rigorous stationarity and differencing workflow, produces accurate monthly CO_2 forecasts with interval estimates (hold-out RMSE 0.431 ppm over 24 months). Pairing it with a SHAP-explained lag model provides exact, auditable feature weights, so stakeholders can see both *what* is forecast and *why*. All code, tables and figures are reproducible with `python project.py`.

References

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